

Part A · Multiple Choice

2 points each

1. **d) flux melting**

p. 213 — the subducting crust does not itself melt, but heat drives water out of its minerals; that water rises into the overlying mantle and lowers its melting temperature. Decompression melting operates at divergent boundaries and mantle plumes.

2. **b) more viscous and richer in volatiles, so gases cannot escape easily**

pp. 215–216 — more silica means more polymerization and higher viscosity. Mafic magmas carry 1–3% volatiles, felsic magmas 4–7%. Trapped gas builds pressure until something breaks.

3. **c) Composite volcano**

p. 219 — Mt. St. Helens is the book's example. Their magma averages intermediate but ranges from felsic to mafic, so eruption style varies widely.

4. **a) their mafic magma has low viscosity and flows a long way before cooling**

p. 220 — the larger ones exceed 100 km across. Mauna Loa is about 200 km in diameter and is the world's largest mountain by volume.

5. **b) a mudflow or debris flow related to a volcano**

p. 224 — most are caused by snow and ice melting during an eruption. The 1985 lahar from Nevado del Ruiz killed 23,000 people in Armero, Colombia, about 50 km from the volcano.

6. **a) a caldera**

p. 220 — Kilauea's caldera is 4 km long and 3 km wide. Its floor rises and falls as the magma chamber below expands and deflates.

Part B · True or False

1 point each

7. **TRUE**

p. 222 — indirect effects, such as famine following climate disruption, account for roughly 8 million deaths in historical times, while direct effects account for fewer than 200,000, about 2.5% of the total.

8. **FALSE**

That describes aa. Pahoehoe is the smooth, ropy texture formed when non-viscous lava flows gently and its skin gels and wrinkles over the still-moving flow beneath. (p. 221)

9. **Cinder cones: small (tens to hundreds of meters), steep, mostly mafic, built from gas-rich fragments. Composite: medium (thousands of meters), moderately steep, felsic to mafic, explosive to effusive. Shield: large (about 1,000 m high and up to 200 km across), very gentle, almost always mafic, typically effusive.**

Rubric: 1 pt each volcano, requiring at least two of the three attributes correct. **Cinder cone** — 10 to a few hundred meters, slopes over 20°, mostly mafic scoria from gas-rich early-stage eruptions, usually monogenetic and easily eroded; example Mt. Lassen. **Composite** — thousands of meters, slopes 10–30°, almost all at subduction zones, magma varying felsic to mafic with correspondingly variable eruption style; example Mt. St. Helens. **Shield** — about 1,000 m high and up to 200 km across, slopes 2–10°, almost always mafic, typically effusive; example Kilauea. (pp. 218–221)

10. **A mantle plume beneath the Pacific Plate feeds volcanoes at a fixed point. The plate moves northwest at about 7 cm per year, carrying each volcano off the plume and leaving a line of progressively older, extinct volcanoes.**

Rubric: 1 pt — a mantle plume, an ascending column of hot rock from deep in the mantle, partially melts near the base of the lithosphere to feed volcanoes at a hotspot. 1 pt — the plume stays roughly fixed while the plate moves, so earlier-formed volcanoes drift away and go extinct while the youngest sits directly over the plume; the current plume lies beneath Mauna Loa, Kilauea, and Loihi. 1 pt — because the ages line up along the direction of motion, geologists read both the *direction* and the *speed* of plate movement from the chain; here the Pacific Plate moves northwest at about 7 cm per year. (pp. 214, 197, 220)